



c23-c-304

23127

BOARD DIPLOMA EXAMINATION, (C-23)

OCTOBER/NOVEMBER—2024

DCE – THIRD SEMESTER EXAMINATION

SURVEYING II

Time : 3 Hours]

[Total Marks : 80

PART—A

3×10=30

- Instructions :** (1) Answer **all** questions.
(2) Each question carries **three** marks.
(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. List the fundamental lines of a theodolite and write its relationship.
2. Define (a) Latitude of a line and (b) Departure of a line.
3. Name the instruments used in trigonometric levelling and state their functions.
4. Write the principle of tachometric surveying. List any two uses of tachometric surveying.
5. A tacheometer has stadia interval 2.4 mm. The focal length of the object glass is 24 cm and the distance from the object glass to the trunnion axis is 10 cm. Calculate its tachometric constants i.e. k and c .
6. Draw a neat sketch of simple curve and name its elements.
7. Define degree of a curve and state the relation between radius and degree of curve of chord or arc length of 30 m.

8. List any three applications of GPS in civil engineering.
9. Briefly write about history of drones.
10. List any three uses of drone surveying in civil engineering.

PART—B

10×5=50

Instructions : (1) Answer *any five* questions.
 (2) Each question carries **ten** marks.
 (3) Answers should be comprehensive and criterion for valuation is the content but not the length of the answer.

11. Explain the method of conducting traverse survey by deflection angle method with neat sketch.
12. The following are the corrected consecutive coordinates of a closed traverse. Calculate the area of the traverse by independent coordinate method.

AB	+77.062	+312.139
BC	+248.421	+101.734
CD	+123.993	-254.686
DE	-197.161	-280.333
EA	-252.315	-121.146

13. Find the evaluation of the top of church spire *M* from the following data :

Instrument at	Sighted to	Vertical angle	Remarks
<i>P</i>	<i>M</i>	+40°20'	Staff reading on BM = 1.600 m
<i>Q</i>	<i>M</i>	+20°20'	Staff reading on BM = 1.255 m

Station *P* and *Q* and the top of the church spire (*M*) are in same vertical plane *RL* Of Bench Mark = +192.765 m and Distance *PQ* = 65 m

- 14.** A tachometer was setup at a station *A* and the following readings were obtained on a staff held vertically. Calculate the horizontal distance *AB* and *RL* of *B*, when the constants of instrument are 100 and 0.15.

Inst. Station	Staff Station	Vertical angle	Hair Reading (m)			Remarks
<i>A</i>	<i>BM</i>	$-6^{\circ}40'$	1.2	1.9	2.6	<i>RL</i> of <i>BM</i>
	<i>B</i>	$+8^{\circ}20'$	0.8	1.6	2.4	= 850.50 m

- 15.** Derive distance and elevation formulas for fixed hair method in tacheometry for inclined sights when the staff held vertical with a neat sketch.
- 16.** Two straights *PQ* and *QR* are connected by a circular curve of 360 m radius. Calculate any five elements of the curve if the deflection angle (φ) is 60° .
- 17.** Two straights meet at an angle of 120° at chainage 2500.80 m. The two straights are connected by a simple circular curve of radius 300 m. Find (a) Length of the curve (b) Chainage at point of commencement and at point of tangency (c) length of long chord (d) apex distance.
- 18.** Write short notes on
- (a) Global Positioning System (GPS)
- (b) Geographical Information System (GIS)

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